

Act III: Casting the Sellers

Production, Costs and Supply Decisions

Module VII: The Shopkeeper's Offer

The Supply Curve



Previously in Marginal Falls

- ◆ Consumers' choices
- ◆ Technology
- ◆ Cost Curves
- ◆ Optimal Production Plan
- ◆ Cost Minimization

Next: Optimal production and Supply curve

Mayor's Price

The Mayor announces the Price of the pastries:

$$P = \$13$$

All pastry sellers must charge the same price.



How much should the sellers produce?

Revenue, Costs and Profits

Total Revenue is the money the firm receives from selling output.

$$\text{TR} = P \times Q$$

Profit is what remains after paying the cost of production.

$$\text{Profit} = \text{TR} - \text{TC}$$

Revenue, Costs and Profits

$$TR = P \times Q$$

$$\text{Profit} = TR - TC$$

Pastries (Q)	Price (P)	Total Revenue (TR)	Total Cost (TC)	Profit
0	\$13		\$500	
50	\$13		\$750	
100	\$13		\$1,100	
150	\$13		\$1,550	
200	\$13		\$2,150	
250	\$13		\$2,900	
300	\$13		\$3,950	

Revenue, Costs and Profits

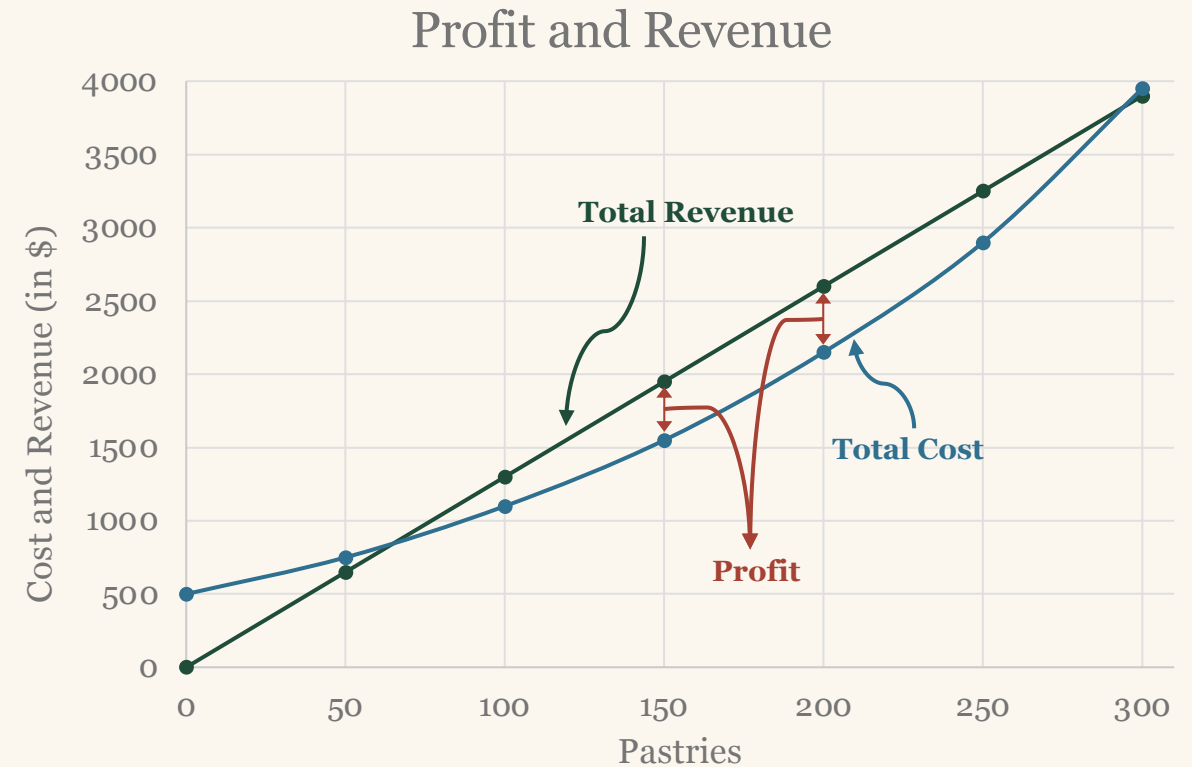
$$TR = P \times Q$$

$$\text{Profit} = TR - TC$$

Pastries (Q)	Price (P)	Total Revenue (TR)	Total Cost (TC)	Profit
0	\$13	\$0	\$500	-\$500
50	\$13	\$650	\$750	-\$100
100	\$13	\$1,300	\$1,100	\$200
150	\$13	\$1,950	\$1,550	\$400
200	\$13	\$2,600	\$2,150	\$450
250	\$13	\$3,250	\$2,900	\$350
300	\$13	\$3,900	\$3,950	-\$50

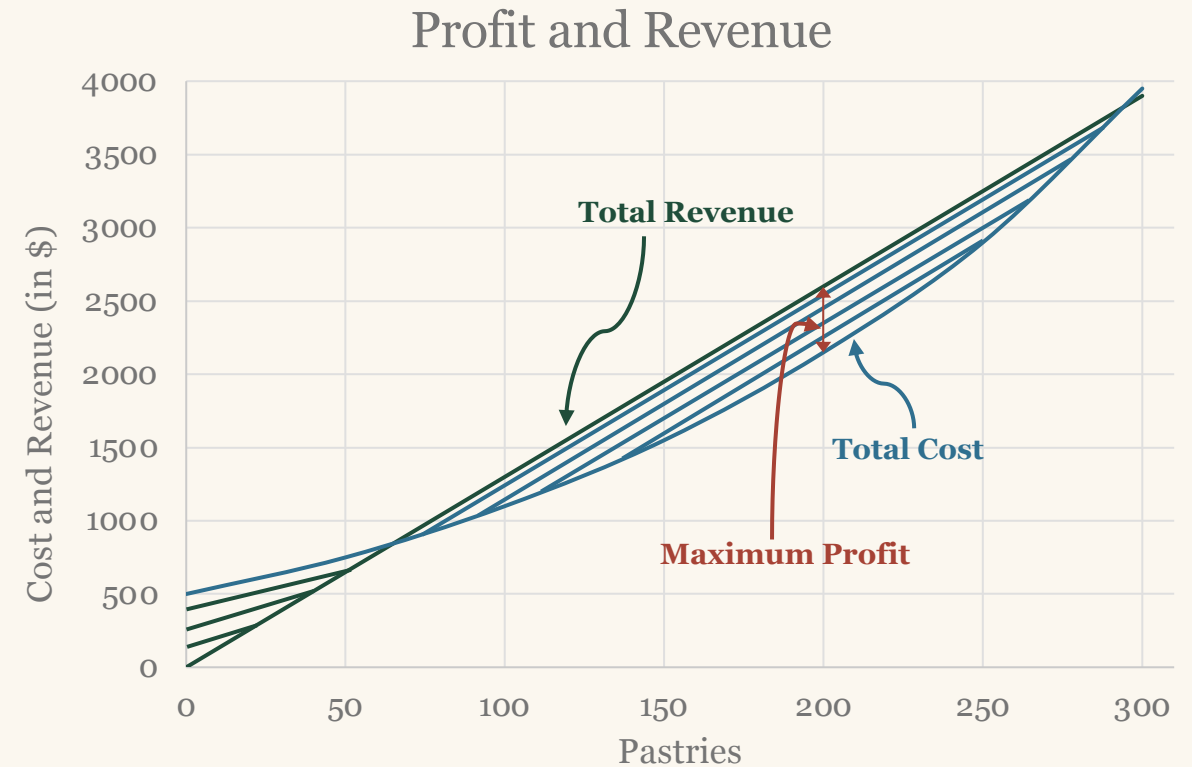
Revenue, Costs and Profits

Pastries (Q)	Price (P)	Total Revenue (TR)	Total Cost (TC)	Profit
0	\$13	\$0	\$500	-\$500
50	\$13	\$650	\$750	-\$100
100	\$13	\$1,300	\$1,100	\$200
150	\$13	\$1,950	\$1,550	\$400
200	\$13	\$2,600	\$2,150	\$450
250	\$13	\$3,250	\$2,900	\$350
300	\$13	\$3,900	\$3,950	-\$50



Revenue, Costs and Profits

Pastries (Q)	Price (P)	Total Revenue (TR)	Total Cost (TC)	Profit
0	\$13	\$0	\$500	-\$500
50	\$13	\$650	\$750	-\$100
100	\$13	\$1,300	\$1,100	\$200
150	\$13	\$1,950	\$1,550	\$400
200	\$13	\$2,600	\$2,150	\$450
250	\$13	\$3,250	\$2,900	\$350
300	\$13	\$3,900	\$3,950	-\$50



One More Pastry

Marginal Revenue is the change in Total Revenue from selling one more unit of Pastry.

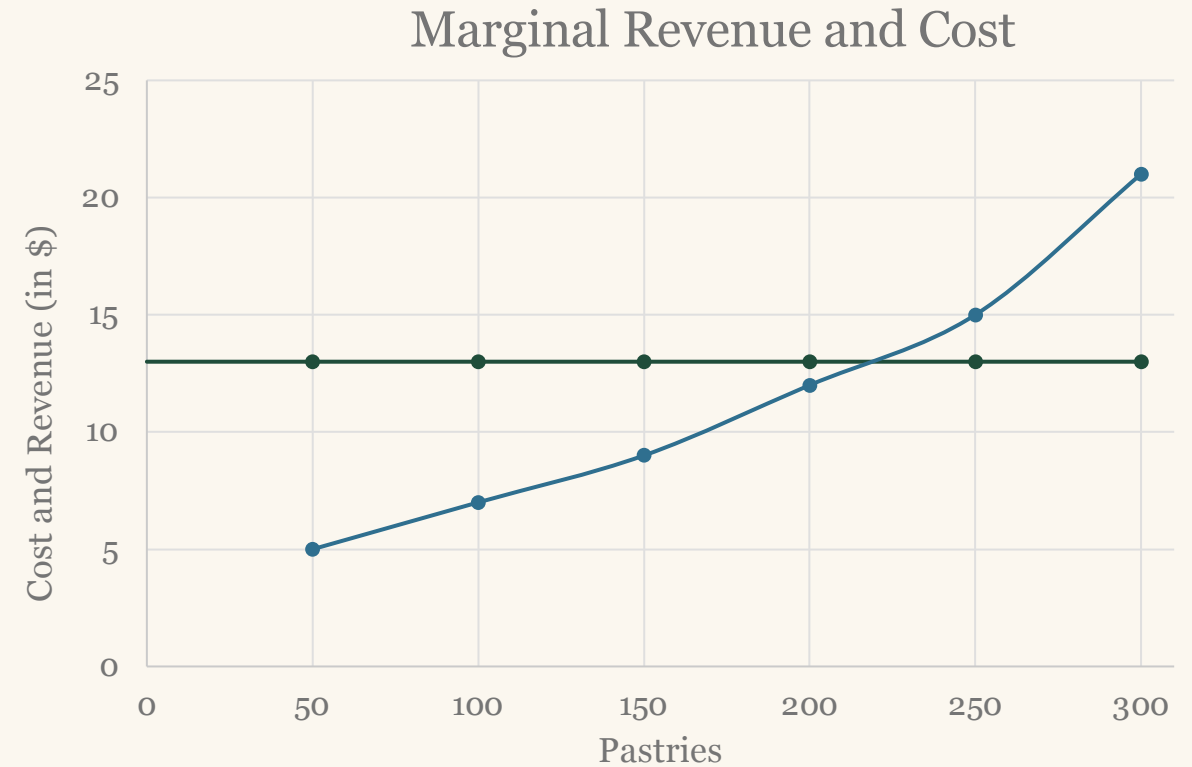
$$MR = \frac{\Delta TR}{\Delta Q}$$

Here, each extra Pastry brings in \$13, because of fixed price.

$$MR = \$13$$

Marginal Revenue

Pastries (Q)	Price (P)	Marginal Revenue (TR)	Total Cost (TC)	Marginal Cost (MC)
0	\$13	-	\$500	-
50	\$13	\$13	\$750	\$5
100	\$13	\$13	\$1,100	\$7
150	\$13	\$13	\$1,550	\$9
200	\$13	\$13	\$2,150	\$12
250	\$13	\$13	\$2,900	\$15
300	\$13	\$13	\$3,950	\$21



Optimal Production

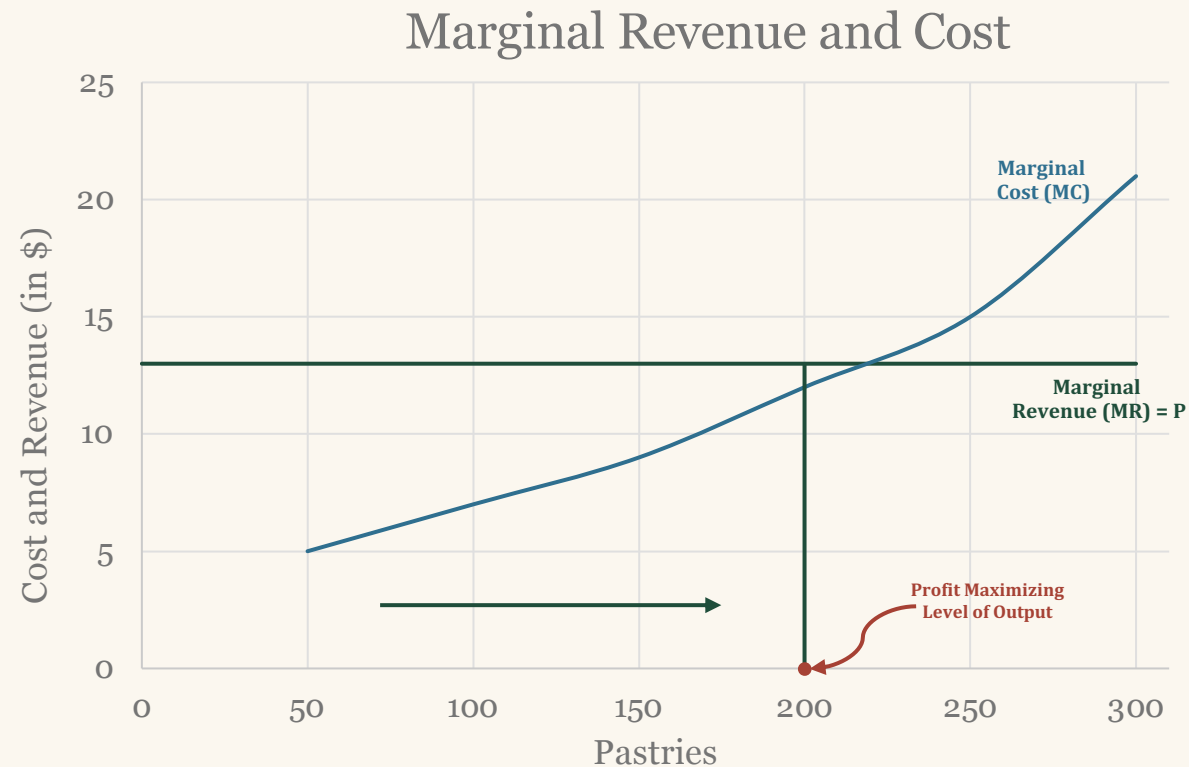
Produce as long as:

$$MR \geq MC$$

Since, $MR = P$

Produce as long as:

$$P \geq MC$$



Optimal Production

Produce as long as:

$$MR \geq MC$$

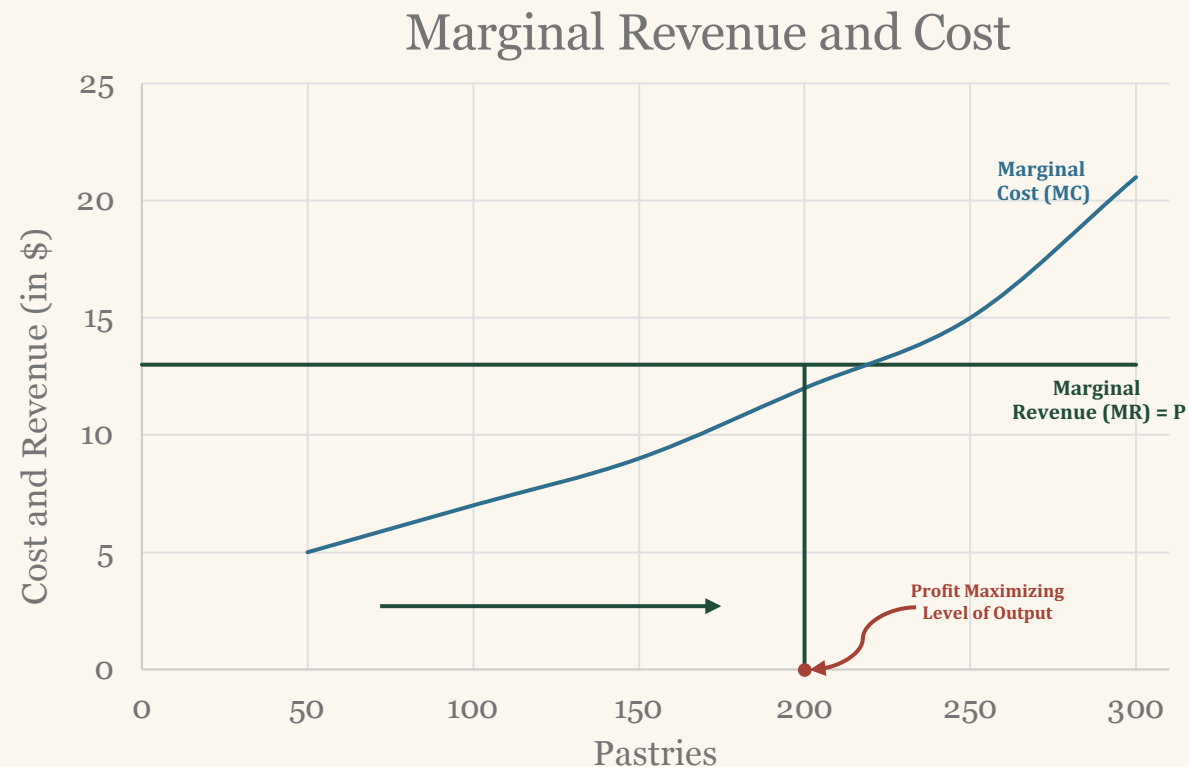
Since, $MR = P$

Produce as long as:

$$P \geq MC$$

Optimal Production,

$$Q^* = 200$$

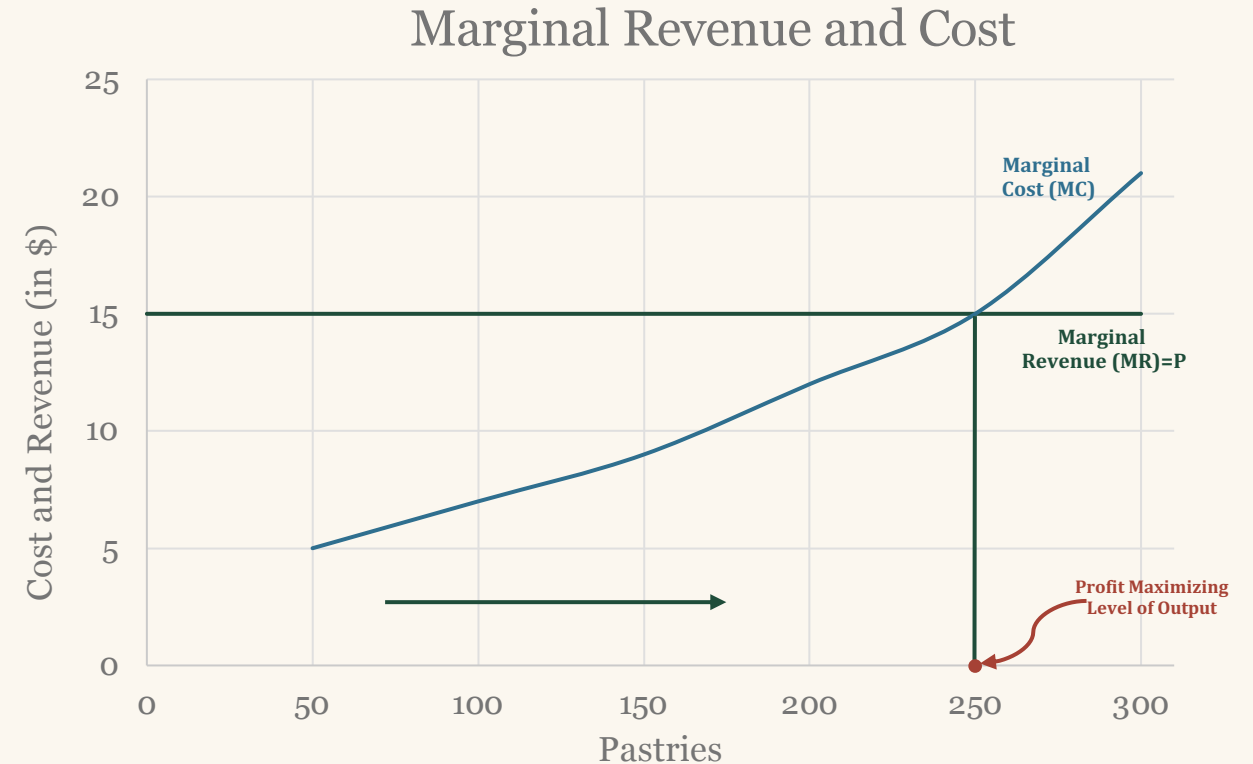


Mayor's New Directive

After the Board Meeting, the
Mayor announces a new
price:

$$P = \$15$$

Optimal Production,
 $Q^* = 250$



What if the Price Keeps Changing?

Mayor's Price	Optimal Quantity
\$5	50
\$7	100
\$9	150
\$12	200
\$15	250
\$21	300

Each price gives a different optimal quantity.

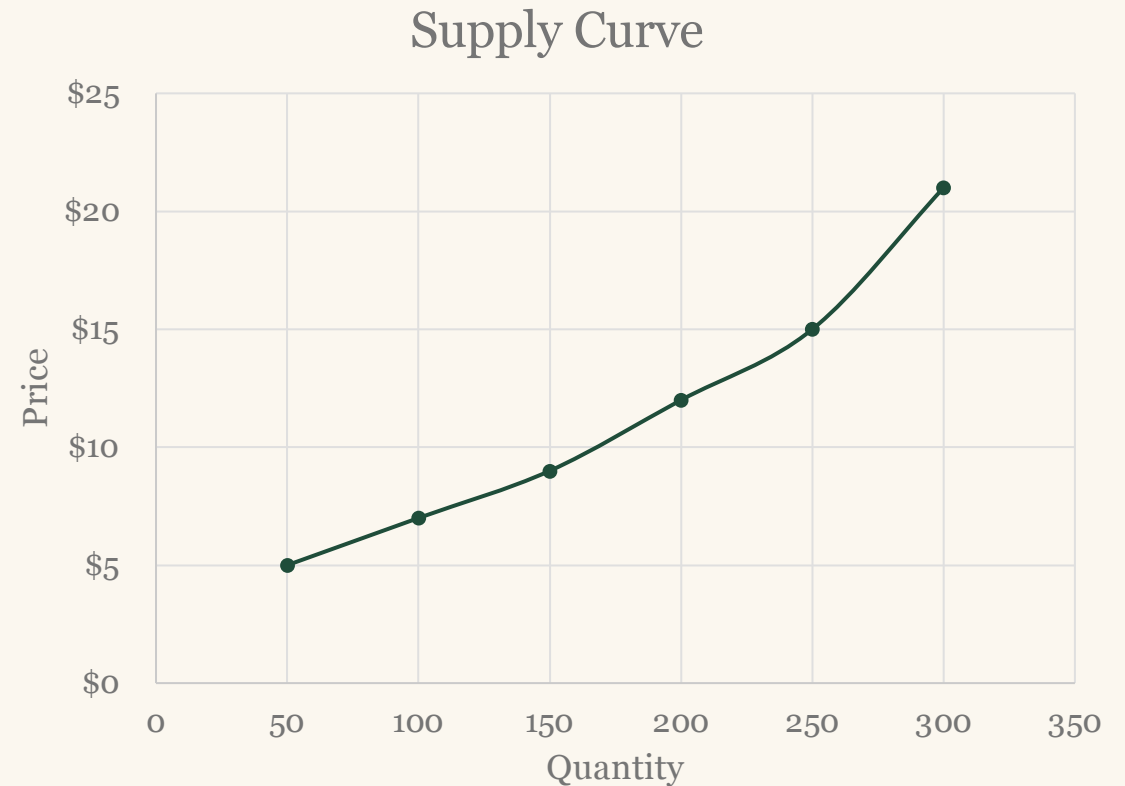
The Baker's Supply Schedule

Price of Pastry	Quantity Supplied
\$5	50
\$7	100
\$9	150
\$12	200
\$15	250
\$21	300

A **supply schedule** is a table showing the relationship between price and quantity supplied.

The Baker's Supply Curve

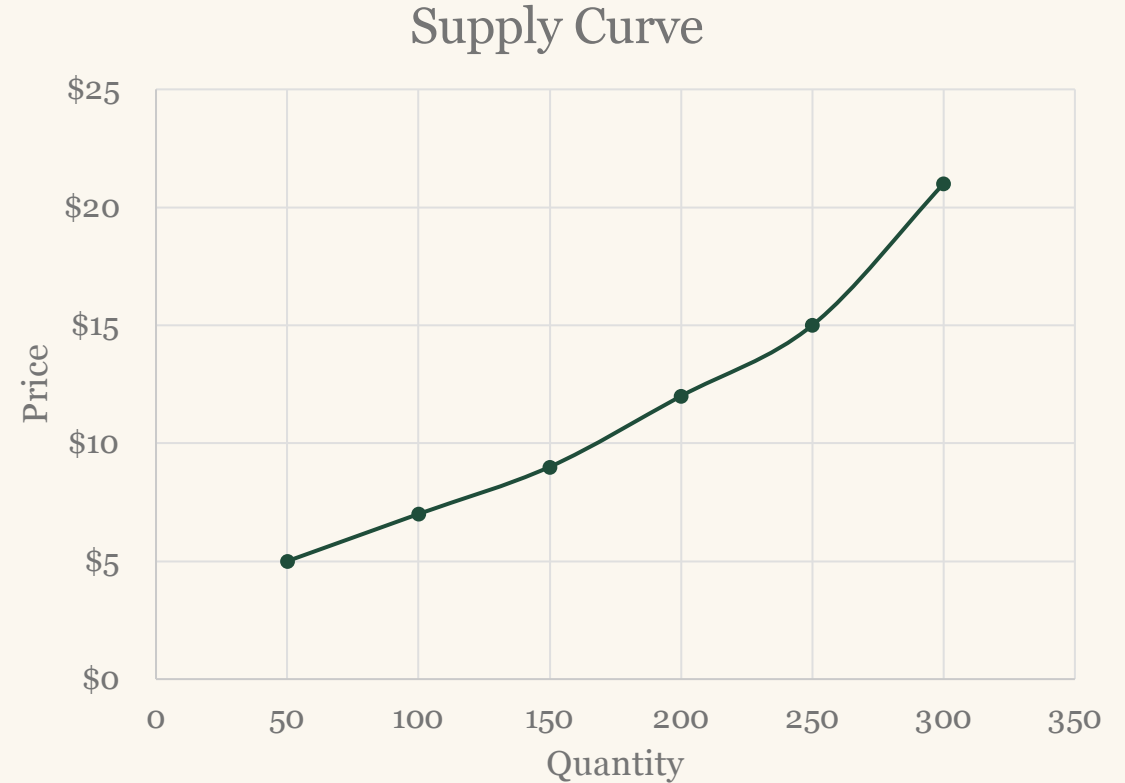
Price of Pastry	Quantity Supplied
\$5	50
\$7	100
\$9	150
\$12	200
\$15	250
\$21	300



The Baker's Supply Curve

A **supply curve** shows the relationship between price and quantity supplied.

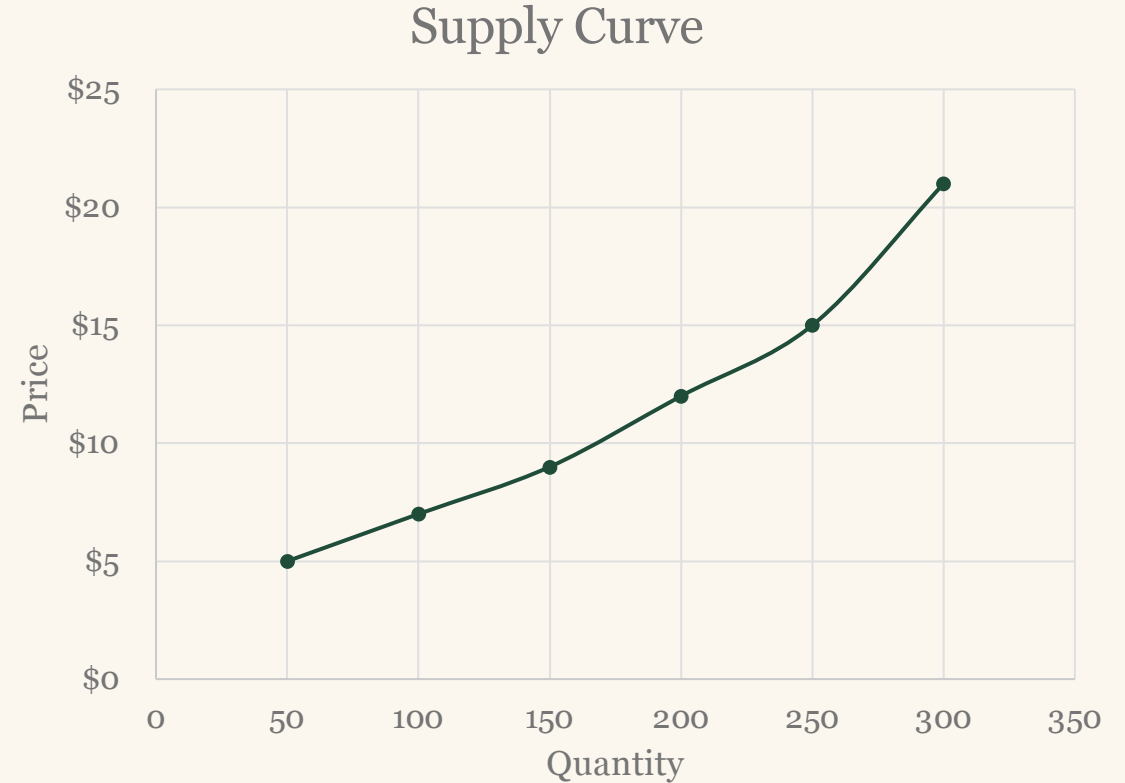
Higher price → Higher quantity supplied.



Law of Supply

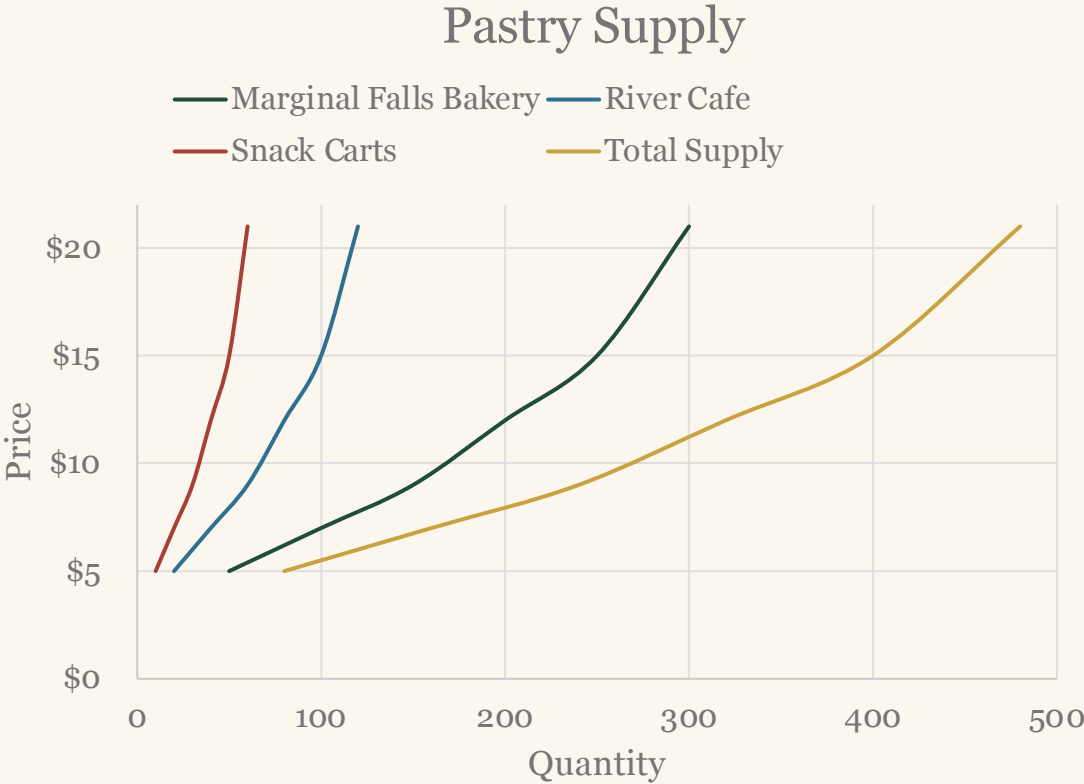
Holding everything else constant
(ceteris paribus):

$$P \uparrow \Rightarrow Q_s \uparrow$$



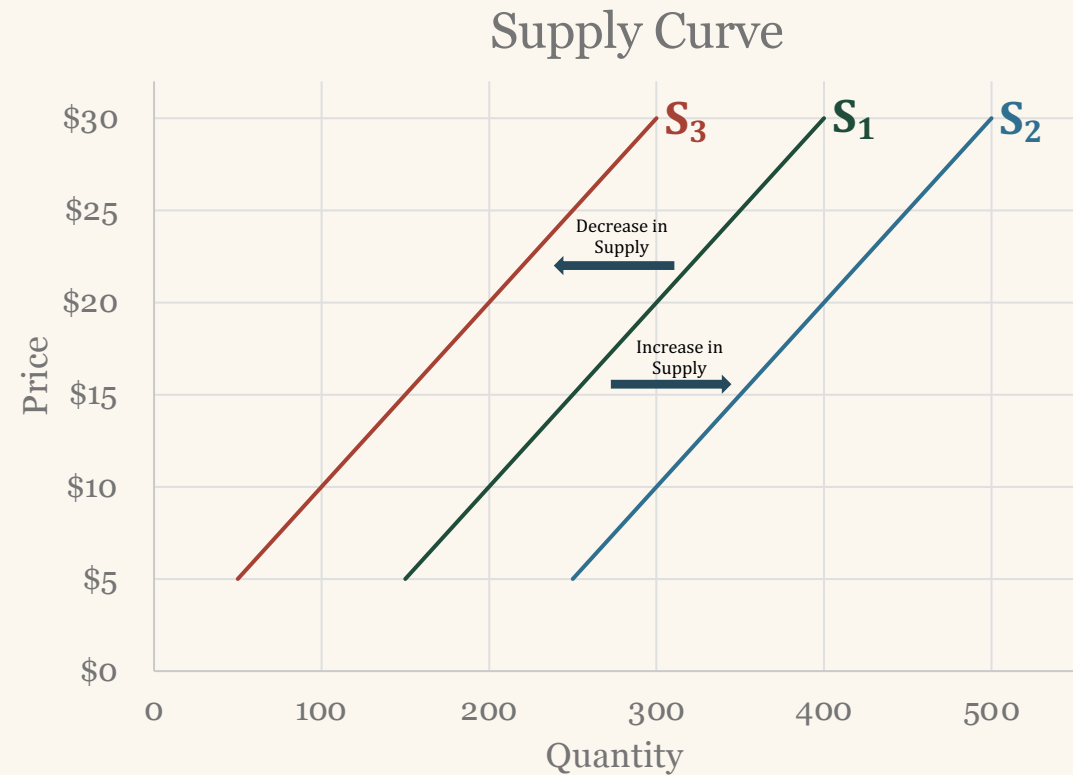
Supply of Pastries in Marginal Falls

Price	Marginal Falls Bakery	River Cafe	Snack Carts	Total Supply
\$5	50	20	10	80
\$7	100	40	20	160
\$9	150	60	30	240
\$12	200	80	40	320
\$15	250	100	50	400
\$21	300	120	60	480



Shifting the Supply Curve

A change in something other than the price of the produced good, that affect supply, shifts the supply curve.



What Shifts Supply?

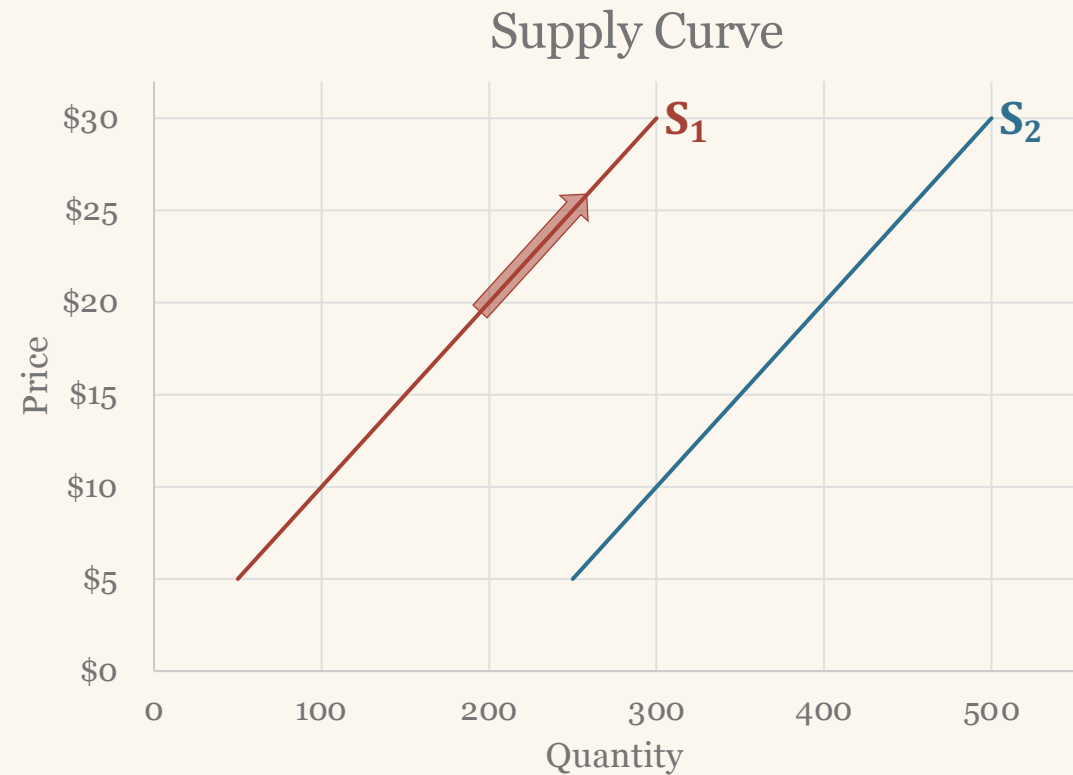
Factor	Example	Supply Effect
Input Prices	Wages or Price of Ovens rise	Supply decreases
Technological Change	Improved Recipe	Supply increases
Natural Disaster/Pandemics	Covid-19	Supply decreases
Prices of Related Goods in Production	Festival Cakes is more profitable during festivals	Supply decreases

Supply Curve Movements

A change in own price,

$P = \$20 \rightarrow \25

Movement along the Supply Curve



Supply Curve Movements

A change in the wage,
 $w = \$30 \rightarrow \15
Shift of the Supply Curve

